CSCE 222: Discrete Structures - Final Exam Reference Guide (Rosen 9th Ed.)

LOGIC & EQUIVALENCES

Operators: $\neg$ (not), $\wedge$ (and), $\vee$ (or), $\oplus$ (xor), $\to$ (if), $\leftrightarrow$ (iff).
Order: $\neg, \wedge, \vee, \to, \leftrightarrow$ (High to Low).
Equivalences:

- $p \to q \equiv \neg p \vee q$
- $\neg(p \wedge q) \equiv \neg p \vee \neg q$ (De Morgan)
- $p \leftrightarrow q \equiv (\neg p \vee q) \wedge (\neg q \vee p)$
- **Identity:** $p \wedge T \equiv p$; $p \vee F \equiv p$
- **Domination:** $p \vee T \equiv T$; $p \wedge F \equiv F$
- **Absorption:** $p \vee (p \wedge q) \equiv p$; $p \wedge (p \vee q) \equiv p$

Fallacies:

- **Affirming Consequent:** $(p \to q) \wedge q \therefore p$ (Error)
- **Denying Antecedent:** $(p \to q) \wedge \neg p \therefore \neg q$ (Error)

RULES OF INFERENCE

$p, p \to q \therefore q$	Modus Ponens
$\neg q, p \to q \therefore \neg p$	Modus Tollens
$p \to q, q \to r \therefore p \to r$	Hypo. Syllogism
$p \vee q, \neg p \therefore q$	Disj. Syllogism
$p \therefore p \vee q$	Addition
$p \wedge q \therefore p$	Simplification
$p, q \therefore p \wedge q$	Conjunction
$p \vee q, \neg p \vee r \therefore q \vee r$	Resolution

PREDICATE LOGIC

Universal: $\forall x P(x)$ (True for all x in domain)
Existential: $\exists x P(x)$ (True for at least one x)
Negation:
$\neg \forall x P(x) \equiv \exists x \neg P(x)$
$\neg \exists x P(x) \equiv \forall x \neg P(x)$

PROOF TECHNIQUES

Direct: Assume p, show q.
Contrapositive: Assume $\neg q$, show $\neg p$.
Contradiction: Assume $p \wedge \neg q$, derive F.
Cases: Prove $(p_1 \vee p_2 \vee \cdots \vee p_n) \to q$ by proving $p_i \to q$ for each i.

SETS & CARDINALITY

$A \subseteq B \iff \forall x(x \in A \to x \in B)$
Operations: $A \cup B$ (or), $A \cap B$ (and), $A - B$ (A not in B).
Complement: $\overline{A} = U - A$.
Power Set: $P(A) = \{S \mid S \subseteq A\}$, $|P(A)| = 2^{|A|}$.
Cartesian: $A \times B = \{(a, b) \mid a \in A, b \in B\}$, $|A \times B| = |A||B|$.
Inclusion-Exclusion: $|A \cup B| = |A| + |B| - |A \cap B|$.

FUNCTIONS

Function: $f : A \to B$. A is domain, B is codomain.
Injective (1-1): $f(a) = f(b) \Rightarrow a = b$.
Surjective (Onto): $\forall y \in B, \exists x \in A, f(x) = y$.
Bijective: Both injective and surjective.
Inverse: f^{-1} exists iff f is a bijection.
Composition: $(f \circ g)(x) = f(g(x))$.

NUMBER THEORY

Divisibility: $a \mid b \iff b = ak$. **Division Alg:** $a = dq + r, 0 \leq r < d$.
GCD/LCM: $\gcd(a, b) \cdot \mathrm{lcm}(a, b) = ab$. **Prime Thm:** $\pi(x) \approx x/\ln x$.
Fermat's Little Thm: $a^{p-1} \equiv 1 \pmod p$ if $p \nmid a$.
Bézout's Thm: $\gcd(a, b) = sa + tb$.
Euclidean Alg: $\gcd(a, b) = \gcd(b, a \mod b)$.
Chinese Remainder: $x \equiv a_i \pmod{m_i}$ has unique soln $\pmod M$.
Modular Inverse: $ax \equiv 1 \pmod m$ exists iff $\gcd(a, m) = 1$.
Wilson's Thm: $(p-1)! \equiv -1 \pmod p$ iff p is prime.

ALGORITHMS & COMPLEXITY

Big-O: $f(n) \leq Cg(n)$ for $n > k$. **Growth:** $1 < \log n < n < n \log n < n^2 < 2^n < n!$.
Master Thm: $T(n) = aT(n/b) + cn^d$:
1. $a < b^d \Rightarrow O(n^d)$ 2. $a = b^d \Rightarrow O(n^d \log n)$ 3. $a > b^d \Rightarrow O(n^{\log_b a})$
Search: Linear $O(n)$, Binary $O(\log n)$. **Sort:** Bubble/Inertion $O(n^2)$.

SEQUENCES & SUMS

Arithmetic: $a_n = a + (n-1)d$. **Sum:** $S_n = \frac{n(2a+(n-1)d)}{2}$.
Geometric: $a_n = ar^{n-1}$. **Sum:** $S_n = \frac{a(r^n-1)}{r-1}$ $(r \neq 1)$.
Series: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$, $\sum i^2 = \frac{n(n+1)(2n+1)}{6}$.

COUNTING

Product (AND): $n_1 \cdot n_2$ **Sum (OR):** $n_1 + n_2$
Incl-Excl: $|A \cup B| = |A| + |B| - |A \cap B|$
Division Rule: n/d (Each item counted d times)
Pigeonhole: N objects, k boxes $\Rightarrow \lceil N/k \rceil$ in one.
 # Functions (n^m): $|B|^{|A|}$ **Injective ($P(n, r)$):** $P(|B|, |A|)$

PERMS & COMBS

Perms ($P(n, r)$): $\frac{n!}{(n-r)!}$ **Combs ($\binom{n}{r}$):** $\frac{n!}{r!(n-r)!}$
w/ Repetition: n^r (Ordered) $\binom{n+r-1}{r}$ (Unordered)
Anagrams/Dist Boxes: $\frac{n!}{n_1!n_2!\ldots n_k!}$.
Indist Objs $\to$ Dist Boxes: $\binom{n+r-1}{r}$ (Stars & Bars).

Identities: $\sum \binom{n}{k} = 2^n$; $\sum (-1)^k \binom{n}{k} = 0$; $\sum 2^k \binom{n}{k} = 3^n$.
Hockey-Stick: $\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$. **Derangements:** $D_n = n! \sum \frac{(-1)^k}{k!}$.
Binomial Thm: $(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$.
Pascal's: $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$ **Vandermonde's:** $\binom{m+n}{r} = \sum \binom{m}{r-k}\binom{n}{k}$.

RELATIONS

$R \subseteq A \times A$:

- **Reflexive:** $\forall a, (a, a) \in R$
- **Symmetric:** $(a, b) \in R \Rightarrow (b, a) \in R$
- **Antisymmetric:** $(a, b), (b, a) \in R \Rightarrow a = b$
- **Transitive:** $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$

Equiv. Relation: Refl, Symm, Trans.
Partial Order (Poset): Refl, Antisymm, Trans.
 # Relations on n elements: 2^{n^2}. **Reflexive:** $2^{n(n-1)}$.

RECURRENCE RELATIONS

Lin. Homog. order k: $a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k}$.
Char. Eq: $r^k - c_1 r^{k-1} - \cdots - c_k = 0$.
1. **Distinct roots** r_1, r_2: $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$.
2. **Repeated root** r_1: $a_n = (\alpha_1 + \alpha_2 n) r_1^n$.
Solving: 1. Find char eq. 2. Find roots. 3. Use IC to solve α_i.

INDUCTION & PROBABILITY

Goal: $\forall n \geq a, P(n)$ **Basis:** Show $P(a)$.
Inductive: Assume $P(k)$ (**I.H.**), show $P(k + 1)$.
Strong: Assume $P(a) \wedge \cdots \wedge P(k)$, show $P(k + 1)$.
Bayes' Thm: $P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}$.
Exp. Value: $E(X) = \sum x_i P(X = x_i)$. **Linearity:** $E(X + Y) = E(X) + E(Y)$.
Bernoulli (k success): $C(n, k)p^k q^{n-k}$ $(q = 1 - p)$.

GRAPH THEORY

Handshaking: $\sum_{v \in V} \deg(v) = 2e$.
Euler's Formula: $n - e + f = 2$ (Connected Planar).
Planar Bound: $e \leq 3n - 6$ (if $n \geq 3$); $e \leq 2n - 4$ (No K_3).
Complete Graph K_n: Edges $e = \frac{n(n-1)}{2}$, Deg $(n - 1)$.
Cycles C_n: Edges $e = n$, Deg 2. **Trees:** $e = n - 1$.
Dirac's Thm: $\deg(v) \geq n/2 \Rightarrow$ Hamilton Circuit.
Ore's Thm: $\deg(u) + \deg(v) \geq n \Rightarrow$ Hamilton Circuit.